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Jam shock relief mechanism for gas spring fastening tool

Abstract

A power tool includes an air spring cylinder, a piston movably positioned within the air spring cylinder along a longitudinal axis of the air spring cylinder and a driver blade fixedly attached to the piston. The power tool includes a lifter that is attached to the piston via an elastic member so as to extend side by side with the driver blade. The elastic member serves as a jam bumper that absorbs jam shock due to abnormal tool operation.

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Background/Summary

FIELD

(1) The present disclosure relates generally to a power tool, and more particularly to a torque protection mechanism for a power tool incorporating an air spring.

BACKGROUND

(2) A nail gun (nailer) is a tool which uses sudden application of a force to drive a nail or other fastener into a workpiece. A variety of mechanisms have been developed to supply the required force including the so-called “air spring”. An air spring uses the compressibility of gas, which may be air, nitrogen, etc., (herein referred to simply as “air”) to store energy which is released to forcefully move a driver which in turn forces the fastener into a workpiece. In particular, a motor is used to force a piston to compress the air within a cylinder. When a user presses a trigger on the nailer, the piston is released and the compressed gas forces the piston to move rapidly along a working axis of the nailer. A driver attached to the piston is thus driven into a fastener thereby driving the fastener into the workpiece.

(3) In many air spring applications, a rack and pinion arrangement is used to compress and release the piston. In these devices a motor drives a pinion gear, and the pinion gear includes teeth extending from the periphery of the pinion gear which engage a rack fixed to the piston thereby forcing the piston to compress the gas. In order to release the piston, a portion of the pinion gear has a “tooth gap” wherein no teeth are provided along the periphery of the pinion gear. Consequently, when a user presses the trigger of the nailer with the pinion gear's last tooth before the tooth gap engaged with the rack, the motor rotates the pinion gear to a position at which the

teeth of the pinion gear no longer engage the rack, allowing the pressure of the gas in the cylinder to move the piston along the working axis.

(4) This type of device is typically configured such that once the pinion gear is rotated by the motor to allow the piston to be moved by the compressed gas, the motor simply continues to rotate the pinion gear for one complete rotation of the pinion gear. Accordingly, the tooth gap of the pinion gear is selected such that the rack is engaged by a first tooth of the pinion gear only after the piston has completed its travel along the working axis. The continued rotation of the motor for one revolution of the pinion gear then drives the piston in the opposite direction along the working axis until the last tooth of the pinion gear before the tooth gap is engaged with the rack thereby compressing the gas with the piston. Thus, the pinion gear is moved in one complete rotation from the initiation of the sequence (pressing of the trigger) until the system is ready for the next pressing of the trigger.

(5) The above-described configuration works very well under normal operating conditions. Problems arrive, however, if the driver/piston do not travel to the designed extent along the working axis under the power of the compressed gas. Such situations can occur, for example, if a nail becomes jammed. In such situations, the motor continues to turn and the pinion gear is rotated for one complete turn. Because the piston is not fully extended along the working axis, however, as the first tooth of the pinion gear is rotated into contact with the rack, the tooth engages the rack at a midpoint of the rack rather than at the end of the rack. Consequently, the piston is fully retracted before the pinion gear has completed one full revolution.

(6) Even though the piston is fully retracted in these situations before one complete rotation of the pinion gear, the motor continues to turn forcing the pinion gear toward a full rotation. The continued rotation of the motor forces the pinion gear teeth to momentarily disengage. Upon the disengagement, the compressed air in the cylinder forces the piston (and hence the rack) along the working axis. At the same time the motor rotates another tooth of the pinion gear into engagement with the rack which is now moving, resulting in a forceful impact between the pinion gear and the rack. Depending upon how much of the piston stroke was initially truncated, this can result in multiple shocks as the pinion gear is rotated until the pinion gear has completed one full rotation and the last tooth of the pinion gear is impacted by the rack.

(7) The forceful collision(s) of the pinion gear and the rack is not only disconcerting to a user, it also creates a torsional shock load which propagates along the drive path from the pinion gear into drive gear of the nailer. The shock load, also referred to as a “jam shock”, can lead to stress fractures within the main drive/gearing of the nailer resulting in catastrophic failure. While it is possible to provide materials which can withstand jam shock, such materials tend to be heavy which increases the weight of the portable tool which is undesired in a portable tool.

(8) Therefore, there is a need to reduce and/or eliminate the shock load of air spring systems.

SUMMARY

(9) One approach for providing a torque protection mechanism for a power tool incorporating an air spring is to provide a shock dampening device in the load path of the power tool that absorbs shock loads such as occur during a nail jam event during power tool use. The effectiveness of this approach is dependent upon where the shock dampening device is positioned within the power tool. In the power tool described and claimed herein, a shock dampening device (e.g., a jam bumper) is positioned within the tool at a location in which in the load path is made as short as possible, whereby the shock load on the components of the tool is reduced and/or eliminated. In particular, the jam bumper is disposed between the lifter and the piston.

(10) By locating the jam bumper between the lifter and the piston, when a jam shock occurs, the piston force is not transmitted to the rack teeth, the lifter gear, or into gears of the gearbox and the gearbox housing.

(11) In some aspects, a power tool includes an air spring cylinder and a piston movably positioned within the air spring cylinder along a longitudinal axis of the air spring cylinder. The power tool

includes a driver blade fixedly attached to the piston, a lifter that is attached to the piston via an elastic member and a lifter gear. The lifter gear includes a lifter gear wheel portion and a plurality of teeth that extend radially from the lifter gear wheel portion and are configured to engage the lifter. The power tool includes a motor including a motor output and a hub. The hub includes a first end operably connected to the motor output and a second end including a hub wheel portion that is fixed to the lifter gear whereby the motor is configured to lift the lifter via the lifter gear.

(12) In some embodiments, the elastic member is disposed between the lifter and the piston.

(13) In some embodiments, the elastic member is annular and is disposed in an annular groove provided in a lifter-facing surface of the piston.

(14) In some embodiments, a first end of the driver blade is fixed to the piston and the elastic member encircles the first end of the driver blade.

(15) In some embodiments, a first end of the lifter is in direct contact with the elastic member.

(16) In some embodiments, an end face of the piston includes a hole, a first end of the driver blade is received within the hole, the elastic member is annular, and the elastic member is disposed in an annular channel that surrounds the blind hole.

(17) In some embodiments, a first end of the lifter has an annular shape that defines a central opening, the central opening encircles the first end of the driver blade, and the first end of the lifter is in direct contact with the elastic member.

(18) In some embodiments, the elastic member has surface features configured to tune the elastic properties of the elastic member.

(19) In some embodiments, a lifter first end is in direct contact with the elastic member and a lifter second end is opposite the first end. The lifter has a lifter first portion that includes the first end, the lifter first portion having a lifter central opening that receives the driver blade therein. In addition, the lifter has a lifter second portion that includes the lifter second end, the lifter second portion being configured to engage with the teeth of the lifter gear.

(20) In some embodiments, the lifter first portion is annular in shape and resides in a plane that is perpendicular to the longitudinal axis of the air spring cylinder.

(21) In some embodiments, the lifter second portion includes a plurality of rollers, the rollers being spaced apart along a longitudinal axis of the lifter.

(22) In some embodiments, the lifter second portion includes a plurality of lifter teeth, the lifter teeth being spaced apart along a longitudinal axis of the lifter.

(23) In some embodiments, the power tool includes a stop bumper that is disposed between the piston and an end of the cylinder, the stop bumper including a stop bumper central opening. The lifter and blade extend through the stop bumper central opening.

(24) In some embodiments, the elastic member is supported by the piston and the lifter includes a lifter first end that is in direct contact with the elastic member. A profile of the lifter first end has a shape and dimensions corresponding to the shape and dimensions of a profile of the elastic member.

(25) In some embodiments, the lifter is placed directly adjacent to the driver blade, allowing for compact packing of nailer components and also allows use of an impact or main bumper having a constant or uniform cross-sectional shape.

(26) As higher kinetic energies are desired by nailer users, larger pistons diameters create larger jam forces, but optimizing lifter profiles and shortening the mechanical jam shock load path with a compressible jam elastomer, as described herein, creates a lighter, better performing nail drive mechanism.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a power tool of the present disclosure with the cap removed and the housing partially removed to show a head valve assembly.
- (2) FIG. 2 is a perspective view of some components of the power tool of FIG. 1 with the housing removed.
- (3) FIG. 3 is a front plan view in partial cross-section of an air spring, lifter gear, lifter, driver blade and piston of the power tool of FIG. 1.
- (4) FIG. 4 is a perspective view of the lifter, piston, lifter gear, planetary gearbox, and motor of the power tool of FIG. 1.
- (5) FIG. 5 is a front plan view of the lifter gear and a portion of the lifter of the power tool of FIG. 1 with the last tooth of the lifter gear engaged with the lifter.
- (6) FIG. 6 is a side cross-sectional view of the hub and lifter gear of the power tool of FIG. 1.
- (7) FIG. 7 is a perspective view of the assembled piston, driver blade, lifter, jam bumper and impact bumper.
- (8) FIG. 8 is a cross-sectional view of the assembled piston, driver blade, lifter, jam bumper and impact bumper of FIG. 7.
- (9) FIG. 9 is a perspective view of the assembled piston, driver blade, lifter and jam bumper with the impact bumper omitted to permit visualization of the jam bumper.
- (10) FIG. 10 is an enlargement of a portion of the cross-sectional view of FIG. 8, showing the jam bumper configuration during normal operation of the tool.
- (11) FIG. 11 is an enlargement of a portion of the cross-sectional view of FIG. 8, showing the jam bumper configuration under a jam shock load as occurs during a nail jam event.
- (12) FIG. 12 is a perspective view of the drive blade.
- (13) FIG. 13 is a perspective view of the lifter.
- (14) FIG. 14 is a perspective view of the jam bumper.
- (15) FIG. 15 is a bottom plan view of the jam bumper.
- (16) FIG. 16 is a cross-sectional view of the assembly of FIG. 9.
- (17) FIG. 17 is a perspective view of the assembled piston, driver blade, alternative embodiment lifter and impact bumper.
- (18) FIG. 18 is a perspective view of the alternative embodiment lifter.
- (19) FIG. 19 is a cross-sectional view of the assembly of FIG. 17.
- (20) FIG. 20 is a perspective view of the WCE extension of FIG. 3.
- (21) FIG. 21 is a perspective view of the cap of FIG. 3.
- (22) FIG. 22 is a perspective view of the flapper valve and plunger of the nailer of FIG. 1.
- (23) FIG. 23 is a partial perspective view of the nailer of FIG. 1 with a portion of the housing and the cap removed to show the location of the head valve assembly.
- (24) FIG. 24 is a partial cross-sectional view of the head valve assembly and air cylinder of the nailer of FIG. 1 with the flapper valve in a non-firing position.
- (25) FIG. 25 is a partial cross-sectional view of the head valve assembly and air cylinder of the nailer of FIG. 1 with the flapper valve in a firing position.
- (26) FIG. 26 is a partial front plan view of the lifter gear and lifter of the power tool of FIG. 1 with the first tooth of the lifter gear engaged with the top roller of the lifter.
- (27) FIG. 27 is a partial front plan view of the lifter gear and lifter of the power tool of FIG. 1 with the first tooth of the lifter gear engaged with the third roller of the lifter.

DESCRIPTION

(28) For the purpose of promoting an understanding of the principles of the disclosure, reference will now be made to the embodiments illustrated in the drawings and described in the following written description. It is to be understood that no limitation to the scope of the disclosure is thereby intended. It is further to be understood that the present disclosure includes any alterations and

modifications to the illustrated embodiments and includes further applications of the principles of the disclosure as would normally occur to one skilled in the art to which this disclosure pertains. (29) Referring to FIG. 1, there is depicted a power tool **100** with an air spring as described below. The power tool in the embodiment of FIG. 1 is a nailer **100**. The nailer **100** includes a housing **102** that defines a drive section **104** and a grip section **106**. A trigger **108** is provided in the grip section **106** and a battery receptacle **110** is configured to removably couple with a battery **112** at the grip section **106**. In other embodiments, the power tool is a corded tool. The nailer further includes a removable nail magazine **114**. A work contact element (WCE) assembly **116** extends out of the housing **102**.

(30) As shown in FIG. 2, within the drive section **104** a cylinder **120** and accumulator **122** are provided. A cap **124** is used to seal the cylinder **120** and the accumulator **122** and defines a headspace **118** above the cylinder **120** and the accumulator **122** (see FIG. 3). A PCBA **126** is operably connected to the trigger **108**, the battery **112**, and a DC brushless motor **128**.

(31) With reference to FIGS. 3-6, a piston **130** is provided within the cylinder **120**. The piston **130** is disposed in the cylinder **120** so as to translate along the cylinder longitudinal axis **226**, also referred to as the drive axis **226**. The piston **130** is prevented from exiting the cylinder second end **123** via an annular, stationary stop bumper **125**. The stop bumper **125** is formed of an elastomeric material and is disposed adjacent the cylinder second end **123**.

(32) A driver blade **180** is fixedly attached to the piston **130**. A lifter **190** extends side-by-side with the driver blade **180** and is connected to the piston **130** via a jam bumper **200**, as discussed in detail below. The lifter **190** includes a number of rollers **194** which are configured to be engaged by teeth **138** of a lifter gear **140**. As shown more clearly in the simplified depiction of FIG. 5, the lifter gear **140**, which functions as a pinion gear, includes a toothed section **142** and a tooth gap section **144**. The tooth gap section **144** is bounded by a first tooth **138F** and a last tooth **138L**.

(33) The lifter gear **140** is operably connected to the motor **128** through a hub **146** (see FIG. 6) and a planetary gearbox **148** (see FIG. 4). The hub **146** is supported by a one-way needle bearing clutch **150**. The hub **146** includes a geared motor side end portion **160** which is operably connected to the planetary gearbox **148**. A body portion **162** of the hub **146** is fixedly connected to an inner race of the one-way needle bearing clutch **150** which is not shown herein in further detail. The body portion **162** is oversized to provide for increased torque capacity with the one-way needle bearing clutch **150**. A wheel portion **164** is positioned at the non-motor facing side of the hub **146**. A central bore **166** extends inwardly from the wheel portion **164** into the body portion **162**. The central bore **166** is provided not only for coupling with the lifter gear **140** as described below, but also to reduce the weight of the hub **146**.

(34) The lifter gear **140** further includes a shaft **154** with an internal bore **156** which lightens the weight of the lifter gear **140**. The shaft **154** is aligned with the central bore **166** and received in the central bore **166**. The lifter gear **140** is secured to the hub wheel portion **164** via bolts **152**. Consequently, the shaft **154** of the lifter gear **140** is maintained within the central bore **166** thereby aligning the hub **146** and the lifter gear **140**.

(35) Referring to FIGS. 3 and 7-8, the nailer **100** includes a fastener driver mechanism. The fastener driver mechanism includes the driver blade **180**, the cylinder **120** and the piston **130**. The piston **130** includes several seals **136** that engage and form a fluid tight seal with an inner surface of the cylinder **120**.

(36) The piston **130** includes a central blind hole **132** provided in the piston lifter-facing surface **131** (e.g., the surface of the piston **130** that faces away from the cap **124**), and a protruding annular boss **134** surrounds the blind hole **132**. The inner surface of the blind hole **132** includes an internal thread **133**. The boss **134** has a non-uniform inner diameter so that a terminal end **135** of the boss **134** has a smaller inner diameter than does the blind hole **132**.

(37) The piston **130** includes an annular recess **137** that surrounds the boss **134**. The recess **137** is shaped and dimensioned to receive a portion of the jam bumper **200** therein. In the illustrated

embodiment, the recess **137** has a rectangular cross-sectional shape and receives the jam bumper **200** in a slip fit or press fit.

(38) Referring to FIGS. **7-9**, **12** and **16**, the driver blade **180** is fixed to the piston **130** so as to protrude from the piston lifter-facing surface **131** and serves as the portion of the fastener driver mechanism that contacts a fastener and drives the fastener into a workpiece. The driver blade **180** is an elongate, solid cylindrical rod having a blade first end **181** that is joined to the piston **130** via, for example, a threaded connection, and a blade second end **182** that is opposed to the blade first end **181**. The driver blade **180** includes a blade longitudinal axis **183** that extends between the blade first and second ends **181**, **182**, and is co-linear with the cylinder longitudinal axis **226**.

(39) The driver blade **180** has a circular cross-section and a diameter that varies along the blade longitudinal axis **183**. In particular, the driver blade **180** includes a blade first portion **186** that includes the blade first end **181** and has a blade first diameter **d1**, and a blade second portion **188** that includes the blade second end **182** and has a blade second diameter **d2** that is smaller than the blade first diameter **d1**.

(40) The driver blade includes an integrally-formed annular protrusion **185** at the transition between the blade first and second portions **186**, **188**. The diameter **d3** of the annular protrusion **185** is greater than the blade first diameter **d1**. A shoulder **185(1)** is formed at the intersection of the blade first portion **186** and the annular protrusion **185**, whereas the transition **185(2)** between the annular protrusion **185** and the blade second portion **188** is smoothly curved. The annular protrusion **185** is truncated on one side of the blade longitudinal axis **183**, providing a discontinuity **185(3)** in the periphery of the annular protrusion **185**.

(41) The blade first portion **186** includes a circumferentially-extending groove **189** that adjoins the shoulder **185(1)** of the annular protrusion **185**. The groove **189** is configured to receive the terminal end **135** of the boss **134** that protrudes from the piston lifter-facing surface **131**.

(42) The blade first end **181** includes an external thread **184** (shown in FIG. **8**) that engages with a corresponding internal thread **133** provided on the inner surface of the piston blind hole **132**. The blade external thread **184** terminates at groove **189**. The shoulder **185(1)** abuts the piston boss **134** when the driver blade **180** is fully engaged with, and secured to, the piston **130**. Since the driver blade **180** is on the same axis **183** as the piston centerline axis **121** and since the driver blade **180** is rigidly coupled to the piston **130**, the maximum impact energy is permitted to be delivered to the fastener during a driving operation of the tool **100**.

(43) The blade second end **182** terminates in a blunt tip **187** that is perpendicular to the blade longitudinal axis **183** and provides a fastener contact surface during a driving operation of the tool **100**.

(44) The driver blade **180** is configured, for example via conventional forming and treating processes, to accommodate the frequent, high-load impacts associated with driving fasteners into substrates (such as wood, concrete, etc.) having a range of hardnesses.

(45) Referring to FIGS. **3**, **8-9** and **16**, the lifter **190** is part of a lift mechanism that returns piston **130** and the driver blade **180** to a position in the cylinder **120** in which the air in the cylinder is compressed and the nailer is ready to drive a fastener. The lift mechanism also includes the lifter gear **140**, the hub **146**, the planetary gearbox **148** and the motor **128**.

(46) Like the driver blade **180**, the lifter **190** is attached to the piston **130**. Unlike the driver blade **180**, the lifter **190** is not fixed relative to the piston **130**. Instead, the lifter **190** is connected to the piston **130** via a jam bumper **200**, described in detail below. The jam bumper **200** permits the lifter **190** to move a limited amount in the axial direction of the driver blade **180** (e.g., a limited amount in a direction parallel to the driver blade longitudinal axis **183**).

(47) The lifter **190** is connected to the piston **130** through the jam bumper **200** so as to protrude from the piston lifter-facing surface **131**. The lifter **190** serves as the portion of the lift mechanism that engages with the lifter gear **140** and drives the piston **130** axially within the cylinder **120**. The lifter **190** is an elongate structure that extends side-by-side with the driver blade **180**. The lifter **190**

has a lifter first end **191** that is joined to the jam bumper **200**, and a lifter second end **192** that is opposed to the lifter first end **191**. The lifter **190** includes a lifter longitudinal axis **193** that extends between the lifter first and second ends **191**, **192**. The lifter longitudinal axis **193** is parallel to, and radially offset with respect to, the blade longitudinal axis **183**.

(48) The lifter **190** has a cross-sectional shape and a dimension that varies along the lifter longitudinal axis **193**. In particular, the lifter **190** includes a lifter first portion **196** that includes the lifter first end **191** and defines an annular body **195** that encircles the driver blade first end **181** and the piston boss **134**. The annular body **195** resides in a plane that is perpendicular to the cylinder longitudinal axis **226** and lifter longitudinal axis **193**, and a central opening **195(1)** of the annular body **195** is centered on the cylinder longitudinal axis **226**. The lifter **190** includes an elongated lifter second portion **198** that includes the lifter second end **192**. The lifter second portion **198** is coextensive with the lifter longitudinal axis **193**. The lifter second portion **198** defines a ladder structure in which the rungs of the ladder correspond to pivot pins of the rollers **194**. The lifter **190** includes a bridging portion **197** that extends between the lifter first and second portions **196**, **198**.

(49) The jam bumper **200** is an annular structure formed of an elastomer such as polyurethane or rubber. The jam bumper **200** has a cap-facing surface **201** and a lifter-facing surface **202** that is opposite the cap-facing surface **201**. The jam bumper **200** has an outer peripheral edge **203** and an inner peripheral edge **204**, the inner peripheral edge **204** defining a central opening **205**. The inner diameter **d4** (e.g., the diameter of the central opening **205**) is large relative to the outer diameter **d5** and the ratio of inner diameter **d4** to outer diameter **d5** is determined by the requirements of the specific application. In the illustrated embodiment, for example, the ratio of inner diameter **d4** to outer diameter **d5** is 0.6. The jam bumper **200** has a rectangular cross-sectional shape. In the illustrated embodiment, the axial dimension of the jam bumper **200** cross-section is slightly greater than the radial dimension.

(50) In the illustrated embodiment, the axial dimension of the jam bumper **200** is sufficient to permit the jam bumper lifter-facing surface **202** to reside outside the recess **137** of the piston **130** (e.g., the lifter-facing surface **202** stands proud relative to the lifter-facing surface **131** of the piston) when the jam bumper is in an unloaded state (FIG. **10**). The unloaded state occurs, for example, during normal operation of the nailer **100**.

(51) In the event of a nail jam (which is an example of abnormal operation of the nailer **100**), one or more forceful collisions of the lifter gear **140** and the lifter **190** may occur. In the event of a forceful collision of the lifter gear **140** and the lifter **190**, the jam bumper **200** is axially compressed between the lifter first end **191** and the piston **130** as shown in FIG. **11**. The momentary compression of the jam bumper **200** and resulting axial movement of the lifter **190** absorbs the “jam shock” with the shortest possible load path. For example, the piston force is reduced by the jam bumper **200**, as the lifter rollers **194** skip relative to the teeth **138** of the lift gear **140**. During such an event, the jam bumper lifter-facing surface **202** may momentarily recede below the lifter-facing surface **131** of the piston **130**.

(52) In some embodiments, the jam bumper **200** may have some level of preload to maintain contact between the lifter annular body **195** and the shoulder **185(1)** of the driver blade **180**. However, if the lifting force compresses the jam bumper **200** excessively, accelerated wear can occur.

(53) Each of the cap-facing surface **201** and the lifter-facing surface **202** may be generally planar and free of ridges, bumps, protrusions or channels. However, in the illustrated embodiment, the lifter-facing surface **202** includes surface features that permit tuning of the elastic properties of the jam bumper **200**. For example, the lifter-facing surface **202** of the jam bumper **200** includes circumferentially spaced recesses **206**, giving the lifter-facing surface a crenellated appearance. By modification of the number, shape and dimensions of the recesses **206**, the elastic properties of the jam bumper **200** can be tuned. This tuning can replace, or be used in combination with, other property-tuning methods such as modification of bumper dimensions, material selection, durometer

selection, etcetera.

(54) By the above-described configuration, the driver blade **180** and the lifter **190** are each attached to the piston **130** but are not fixed to each other. Instead, the lifter **190** moves independently of the driver blade **180** and is axially moveable relative to both the piston **130** and the driver blade **180** to an extent permitted by the elasticity of the jam bumper **200**.

(55) Referring to FIGS. **17-19**, the above-described embodiment includes the lifter **290** that employs rollers **194** to engage the teeth **138** of the lifter gear **140**. The power tool **100** is not limited to a lifter having rollers. For example, the power tool **100** may employ an alternative embodiment lifter **290** that employs teeth **294** to engage the teeth **138** of the lifter gear **140**.

(56) Like the lifter **190** described above, the alternative embodiment lifter **290** is connected to the piston **130** via the jam bumper **200** so as to protrude from the piston lifter-facing surface **131**. The lifter **290** is an elongate structure that extends side-by-side with the driver blade **180**. The lifter **290** has a lifter first end **291** that is joined to the jam bumper **200**, and a lifter second end **292** that is opposed to the lifter first end **291**. The lifter **290** includes a lifter longitudinal axis **293** that extends between the lifter first and second ends **291, 292**. The lifter longitudinal axis **293** is parallel to, and radially offset with respect to, the blade longitudinal axis **183**.

(57) The lifter **290** has a cross-sectional shape and a dimension that varies along the lifter longitudinal axis **293**. In particular, the lifter **290** includes a lifter first portion **296** that includes the lifter first end **291** and defines an annular body **295** that encircles the driver blade first end **181** and the piston boss **234**. The annular body **295** resides in a plane that is perpendicular to the cylinder longitudinal axis **221** and lifter longitudinal axis **293**, and a central opening **295(1)** of the annular body **295** is centered on the cylinder longitudinal axis **221**. The lifter **290** includes an elongated lifter second portion **298** that includes the lifter second end **292**. The lifter second portion **298** is coextensive with the lifter longitudinal axis **293**. The lifter second portion **298** defines rack including rack teeth **294** that protrude radially in a direction away from the drive blade **180**. The lifter **290** includes a bridging portion **297** that extends between the lifter first and second portions **296, 298**. The bridging portion **297** is angled relative to the lifter longitudinal axis **293** so that the lifter second portion **298** at least partially underlies the central opening **295(1)**.

(58) The lifter **190**, which contains a plurality of rollers **194** or teeth **294** is parallel to the driver blade **180** and is offset from the blade longitudinal axis **183** only by the distance required to create a structural column. In some embodiments, the lifter **190** is in direct contact with the driver blade **180**, with the driver blade **180** complementing the operating loads of the lifter **190**. It is understood that a friction reduction element could be placed between the fixed driver blade **180** and the lifter **190**, if the nailer application has a high likelihood of jam shock.

(59) In the above-described configuration, the jam bumper **200** accommodates jam shock in the tool **100**. For this reason, both the lifter **190** and driver blade **180** can be optimized for their primary function rather than to accommodate jam shocks. For example, since the tool **100** includes the jam bumper **200** as described, the driver blade **180** may be optimized for fastener impact, while the lifter **190** may be optimized for wear, toughness and column loading. In addition, since the lifter **190** and the driver blade **180** are not integrated and instead are separate structures, only the worn part can be replaced, lowering the total cost of ownership for the end user.

(60) Returning to FIG. **3**, the WCE assembly **116** includes a nose piece **210**, which in this embodiment is the WCE, that is fixedly attached to a WCE stamping **212**. A WCE extension **214**, also shown in FIG. **20**, is attached to the WCE stamping **212** at one end and at the other end includes a bearing portion **216**. The WCE extension **214** further includes shoulders **218**. The WCE extension **214** is maintained in alignment with a plunger **220** by a pair of guides **222** (also shown in FIG. **2**). A WCE spring **224** biases the WCE stamping **212** along the drive axis **226** in a direction away from the WCE extension **214**. The shoulders **218** of the WCE extension **214** act as stops with the lower of the two guides **222** to limit downward travel of the nose piece/WCE **210**, WCE stamping **212**, and WCE extension **214**.

(61) As used, herein, “downward” refers to the direction in which a nail (not shown) is driven by the nailer **100** along the drive axis **226**, which is in the downward direction in the configuration depicted in FIG. **3**. Additionally, for ease of discussion, “movement” of the various components is described herein with reference to the housing **102** of the nailer **100**. In particular, under normal operating conditions the WCE **210**, the WCE stamping **212**, and the WCE extension **214** do not actually move since the WCE **210** is positioned against a work piece. Rather the rest of the nailer **100** is moved to compress the WCE spring **224**. Nonetheless, the WCE **210**, the WCE stamping **212**, and the WCE extension **214**, along with other components, will be described as “moving” for ease of discussion, it being understood that the “movement” simply refers to movement relative to the housing **102**.

(62) Returning to FIG. **1**, a portion of the housing **102** is removed as is the cap **124** (see FIGS. **2** and **21**) to reveal a head valve assembly **238** which is also shown in FIGS. **22-24**. The head valve assembly **238** includes a flapper valve **240** which has a seal **242**, the plunger **220**, and a pivot **244**. The pivot **244** includes a circular pin **246** that fits within an oval pivot bore **248** of the flapper valve **240**. The flapper valve **240**, which can seal the headspace **118**, and thus the accumulator **122**, from the cylinder **120**, includes a pair of fingers **250** that receive a neck portion **252** of the plunger **220**.

(63) The neck portion **252** is located between a head **254** and shoulder **256** of the plunger **220**. The neck portion **252** is configured to slide between the fingers **250** from the side (i.e., in a direction orthogonal to the drive axis **226**), while the head **254** and the shoulder **256** are sized to not pass through the fingers **250** in directions along the drive axis **226**. In some embodiments the neck portion is circular in cross section. In other embodiments the neck portion is configured to allow insertion into the fingers in one orientation, while preventing insertion (or removal) when rotated to a different orientation.

(64) A shaft portion **258** of the plunger **220** extends outwardly of the headspace **118** in an airtight but slidable manner through an insert **260**. The shoulder **256** of the plunger **220** is configured to abut the insert **260**, which is fixedly positioned in the nailer **100**, in a non-firing configuration as depicted in FIG. **24**.

(65) Operation of the nailer **100** is described with initial reference to FIG. **24**. In the configuration of FIG. **24**, the piston **130** is at its full upward position within the air cylinder **120**, and is held at this position by the last tooth **138L** of the lifter gear **140** (see FIG. **5**). In this configuration the air within the upper portion of the air cylinder **120**, the headspace **118**, and the air accumulator **122** is fully pressurized. The pressure differential between the headspace **118** and atmosphere acts across the plunger **220** biasing the plunger **220** downwardly along the drive axis **226** thereby forcing the shoulder **256** of the plunger **220** against the insert **260**.

(66) Because the head **254** of the plunger is larger than the opening defined by the fingers **250** of the flapper valve **240** (in a plane orthogonal to the drive axis **226**), the flapper valve **240** is maintained in a non-firing position, and hence the seal **242**, is held firmly against the upper portion of the air cylinder **120** thus sealing the air cylinder **120** from the headspace **118**. In some embodiments, the pivot bore **248** is circular, which creates a tight seal around the entire circumference of the seal **242**. In the embodiment of FIG. **24**, the pivot bore **248** is oval with the major axis extending along the drive axis **226** and positioned to have the pin **246** centrally located when the shoulder **256** is resting against the insert **260**. Consequently, the force of the seal **242** against the air cylinder **120** is reduced at locations proximate the pivot **244**. The reduced force reduces frictional forces introduced between the seal **242** and the air cylinder **120** which must be overcome when actuating the WCE assembly, allowing the WCE actuating force (described below) to be dominated by forces from the WCE spring **224** and forces resulting from the pressurized air in the headspace acting against the plunger **220** as discussed in further detail below.

(67) The reduced force of the seal **242** against the air cylinder **120** may result in some initial leakage past the seal **242** in the event the air in the headspace **118** is at a higher pressure than the air in the air cylinder **120**, but such leakage does not significantly affect the safety performance of the

head valve assembly **238**. In particular, in the event the piston **130** is inadvertently released from the last tooth **138L**, for example, due to a mechanical or electrical fault, the compressed air in the volume of the air cylinder **120** above the piston **130** will force the piston **130** to begin to move downwardly. The area in the air cylinder **120** above the piston thus depressurizes rapidly.

(68) The pressure in the headspace **118** does not, however, depressurize as rapidly (if at all) since the flapper valve **240** is in a non-firing position which hinders passage of air from the headspace **118** to the air cylinder **120**. Thus, the pressure differential across the flapper valve **240** quickly fully seals the flapper valve **240** even if some leakage initially occurs. Thus, the air in the headspace **118**, and the air in the air accumulator **122** is not allowed to pass freely into the air cylinder **120**.

Accordingly, the piston **130** is driven with a substantially lesser force than during normal operation. This safety feature is provided by flapper valves which are initially tightly seated, flapper valves which are initially not tightly seated, and flapper valves which allow some leakage even when tightly seated. In all instances, because the passage of air into the air cylinder is obstructed, the force applied to a fastener is substantially reduced in the event of an inadvertent firing of the nailer **100**.

(69) Continuing with the description of normal operation of the nailer **100**, with the piston **130** and flapper valve **240** in the configuration of FIG. **24**, a user presses the WCE/nosepiece **210** (see FIG. **3**) against a workpiece (not shown) thereby compressing the WCE spring **224** as the WCE stamping **212** and WCE extension **214** move upwardly, with respect to the housing **102**, along the drive axis **226**. This movement continues until the bearing portion **216** of the WCE extension **214** contacts the lower end of the shaft **258** of the plunger **220**. At this point, additional force must be applied to provide continued upward movement of the WCE **210**, WCE stamping **212**, WCE extension **214**, and plunger **220**.

(70) Specifically, the force required to move the WCE **210** is referred to as the “WCE actuation force”. The WCE actuation force is a design choice which takes into account the weight of the tool and provides a safety factor to ensure the operator is actively pressing the WCE against a workpiece to prevent inadvertent firing of the nailer **100**. In some instances, the WCE actuation force is desired to be the amount of force provided by the tool (the weight of the tool at the nose of the tool) plus about 50% of the total weight of the tool. Thus, for a power tool of 10 pounds with an even weight distribution between the nose and the rear of the tool, the force provided by the tool is about 5 pounds force and the additional 50% requires another 5 pounds force for a total of 10 pounds force.

(71) With respect to the nailer **100**, the WCE actuation force is initially established primarily by the counter force of the WCE spring **224** with some negligible friction forces, and is thus a function of the spring constant of the WCE spring **224**. Thus, the WCE actuation force is initially simply the force needed to overcome the WCE counter-force of the WCE spring **224**. Once the bearing portion **216** contacts the plunger **220**, however, the force of the pressurized air in the headspace **118** against the plunger **220** must also be overcome. This force is a function of the pressure in the headspace **118** along with the diameter of the plunger. By forming the pivot bore **248** as an oval as described above, frictional forces associated with the seal **242** and air cylinder **120** are significantly reduced. Moreover, because the frictional forces between the seal **24** and the air cylinder **120** are significantly reduced, moving the flapper valve **240** does not introduce significant torque on the plunger **220**, thereby minimizing friction associated with movement of the plunger **220**.

(72) Therefore, since the pressure in the head valve is a design parameter which is determined based upon the force needed to drive the fastener, the main determinants of the actuation counterforce are the spring constant of the WCE spring **224** and the diameter of the of the plunger **220**.

(73) Thus, the WCE spring **224** spring constant and the diameter of the plunger **220** can be selected to provide a desired WCE actuation force profile. In one embodiment, the spring constant and the plunger diameter are selected such that the WCE spring **224** and movement of the plunger **220** each account for about 50% of the actuation counterforce as the flapper valve **240** moves into a firing

position. In other embodiments, different actuation counter-force profiles are provided.

(74) Continued application of the WCE actuation force moves the plunger **220** to a firing position as depicted in FIG. **25**. In the configuration of FIG. **25**, a continuous air path is provided between the air accumulator **122** and the air cylinder **120** through the headspace **118**. As shown in FIG. **25**, the opening defined by the fingers **250** is larger than the diameter of the neck portion **252**, allowing the flapper valve **240** to pivot about the pivot pin **246** without torquing the plunger **220** and/or creating significant friction.

(75) A sensor (not shown, typically a Hall sensor) senses the position of the WCE **210**, either directly or indirectly, such as by sensing the WCE stamping **212** or the WCE extension **214** and sends a signal to the PCBA **126** indicating that the WCE **210** has been depressed sufficiently to allow for firing of the nailer **100**. A signal indicating depression of the trigger is also sent to the PCBA **126**. With the flapper valve in the firing position and the trigger depressed, the PCBA **126** “fires” the nailer by energizing the motor **128** thereby rotating the hub **146** in the direction of the arrow **276** in FIG. **3**.

(76) As the lifter gear **140** rotates in the direction of the arrow **276**, the last tooth **138L** is forced out of engagement with the bottom roller **194** in the lifter **190** allowing compressed air entrapped in the cylinder **120** above the piston **130**, as well as compressed air in the headspace **118** and accumulator **122**, to expand thereby forcing the piston **130** along the drive axis **226**. The driver blade **180** is then forced against a nail (not shown) forcing the nail into a workpiece (not shown).

(77) Once the driver blade **180** has been fully extended, the motor **128** will have rotated the lifter gear **140** so that the first tooth **138F** is positioned to engage the first (top) roller as shown in FIG. **26**. Continued rotation of the motor **128** results in continued rotation of the lifter gear **140** resulting in the piston **130**, and hence the driver blade **180**, being lifted to the ready position shown in FIG. **3** by time the motor **128** effects one complete rotation of the lifter gear **140**.

(78) In the event the driver blade **180** does not fully extend, resulting in the configuration of FIG. **27**, then the first tooth **138F** will engage a roller **194** other than the first (top) roller **194**. In FIG. **27**, the first tooth **138F** is shown engaging the third roller **194**. Continued rotation of the motor **128** in this scenario results in continued rotation of the lifter gear **140** resulting in the piston **130**, and hence the driver blade **180**, being lifted to the ready position before the motor **128** effects one complete rotation of the lifter gear **140**. Consequently, jam shock will occur as the motor **128** continues rotating the lifter gear **140** with the piston **130** at the ready position shown in FIG. **3**.

(79) In particular, as the motor **128** continues rotating the lifter gear **140** with the piston **130** at the ready position, the teeth **138** are forced out of engagement with the lifter **190**. The flapper valve **240** will still be in the firing position, accordingly, the air in the accumulator **122** is not yet isolated from the air in the cylinder **120**. Thus, the compressed air in the cylinder **120**, the headspace **118**, and the accumulator **122** will force the piston **130**, and hence the lifter **190**, along the drive axis **226** as a following tooth **138** rotates into the path of a roller **194** of the lifter **190**.

(80) A portion of the force of the impact of the engagement of the tooth **138** with a roller **194** of the moving lifter **190** is transferred to the jam bumper **200** through the contacting portions of the lifter first end **191**. The jam bumper **200** thus absorbs at least a portion of the force of the impact.

(81) Once the last tooth **138L** has engaged the lowest roller, rotation of the motor **128** is stopped. Upon lifting the nailer **100** from the workpiece, the WCE spring **224** forces the WCE **210**, the WCE stamping **212**, and the WCE extension **214** downwardly along the drive axis **226** until the shoulders **218** of the WCE extension **214** contact the lower guide **222**.

(82) The downward movement of the WCE extension **214** allows the compressed air within the headspace **118** to force the plunger **220** outwardly from the headspace **118** in a downward direction along the drive axis **226**. The plunger **220** continues to move along the drive axis **226** until the shoulder **256** once again contacts the insert **260**. As the plunger **220** moves downwardly, the head **254** contacts the fingers **250** and forces the flapper valve **240** to move from the firing position of FIG. **25** to the non-firing position of FIG. **24**. The nailer **100** is thus configured for a subsequent

firing operation.

(83) While the disclosure has been illustrated and described in detail in the drawings and foregoing description, the same should be considered as illustrative and not restrictive in character. It is understood that only the preferred embodiments have been presented and that all changes, modifications and further applications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A power tool, comprising: an air spring cylinder; a piston movably positioned within the air spring cylinder along a longitudinal axis of the air spring cylinder; a driver blade fixedly attached to the piston; a lifter that is attached to the piston via an elastic member; a lifter gear including a lifter gear wheel portion, and a plurality of teeth extending radially from the lifter gear wheel portion and configured to engage the lifter; a motor including a motor output; and a hub including a first end operably connected to the motor output and a second end including a hub wheel portion that is fixed to the lifter gear whereby the motor is configured to lift the lifter via the lifter gear.
2. The power tool of claim 1, wherein the elastic member is disposed between the lifter and the piston.
3. The power tool of claim 1, wherein the elastic member is annular and is disposed in an annular groove provided in a lifter-facing surface of the piston.
4. The power tool of claim 1, wherein a first end of the driver blade is fixed to the piston, and the elastic member encircles the first end of the driver blade.
5. The power tool of claim 4, wherein a first end of the lifter is in direct contact with the elastic member.
6. The power tool of claim 1, wherein an end face of the piston includes a hole, a first end of the driver blade is received within the hole, the elastic member is annular, and the elastic member is disposed in an annular channel that surrounds the blind hole.
7. The power tool of claim 6, wherein a first end of the lifter has an annular shape that defines a central opening, the central opening encircles the first end of the driver blade, and the first end of the lifter is in direct contact with the elastic member.
8. The power tool of claim 1, wherein the elastic member has surface features configured to tune the elastic properties of the elastic member.
9. The power tool of claim 1, wherein the lifter includes a lifter first end that is in direct contact with the elastic member, a lifter second end opposite the first end, a lifter first portion that includes the first end, the lifter first portion having a lifter central opening that receives the driver blade therein, and a lifter second portion that includes the lifter second end, the lifter second portion being configured to engage with the teeth of the lifter gear.
10. The power tool of claim 9, wherein the lifter first portion is annular in shape and resides in a plane that is perpendicular to the longitudinal axis of the air spring cylinder.
11. The power tool of claim 9, wherein the lifter second portion includes a plurality of rollers, the rollers being spaced apart along a longitudinal axis of the lifter.
12. The power tool of claim 9, wherein the lifter second portion includes a plurality of lifter teeth, the lifter teeth being spaced apart along a longitudinal axis of the lifter.
13. The power tool of claim 9, comprising a stop bumper that is disposed between the piston and an end of the cylinder, the stop bumper including a stop bumper central opening, wherein the lifter and blade extend through the stop bumper central opening.
14. The power tool of claim 1, comprising a stop bumper that is disposed between the piston and an end of the cylinder, the stop bumper including a stop bumper central opening, wherein the lifter and blade extend through the stop bumper central opening.
15. The power tool of claim 1, wherein the elastic member is supported by the piston, and the lifter

includes a lifter first end that is in direct contact with the elastic member, a profile of the lifter first end having a shape and dimensions corresponding to the shape and dimensions of a profile of the elastic member.
